XENI ON STELLAR · TECHNICAL ARCHITECTURE DOCUMENT

# Technical Architecture

Migrating Xeni's agentic travel-payment stack from USDC on Hedera to USDC on Stellar, and moving the full 450,000+ wallet base to production. This document covers what exists today, why Stellar fits, and the target design. Delivery staging lives in the companion *Project Plan*.

**From** USDC on Hedera | **To** USDC on Stellar | **Agent rail** x402 + SEP-41 allowance | **Scale** 450K+ wallets

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## 1 · CONTEXT — WHAT'S ALREADY BUILT (HEDERA)

Xeni's AI travel agent already pays for bookings in USDC, in two modes (a human authorizing a chosen deal, or the agent auto-selecting within a signed allowance) and with a hardened refund path — serving a 450,000+ wallet user base. The migration retargets this stack to Stellar.

AGENT PAYMENTS

### AgentService + H-MCP

The traveler's **own wallet** pays treasury in USDC; the agent is the named fee-payer. All chain calls go through one MCP boundary (`hedera-agent-kit`) — the agent never touches the SDK.

SETTLEMENT

### XeniPay

Treasury → supplier and commission payouts in USDC, behind a chain-agnostic `int64` money model and a `chains.BlockchainClient` seam.

MONEY INVARIANTS (CARRIED OVER TO STELLAR)

- → The agent **never holds user funds** — USDC stays in the traveler's wallet until the treasury transfer.
- → The user **never pays the gas token** — the agent account is the named fee-payer.
- → Agent spend is **bounded by a signed allowance**, enforced server-side.
- → Every payment and refund is **audited**, and the money path is idempotent (no double-pay).

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2 · WHY STELLAR — THE PRIMITIVES MAP 1:1

Each mechanism hand-built on Hedera has a native Stellar counterpart, so the migration leans on the platform instead of fighting it.

NAMED FEE-PAYER → FEE-BUMP

### User holds no XLM

Stellar fee-bump transactions let the agent account pay the fee for the user's payment — exactly our "user pays no HBAR" model, as a first-class transaction type.

ON-CHAIN ALLOWANCE → SEP-41

### One signature, then autonomous

A single on-chain `user→agent` allowance is a **Soroban SEP-41 approve**; the agent then pays via `transfer_from` with no second wallet popup — the same path x402 settles on.

AGENT RAIL → NATIVE X402

### A facilitator already exists

The "Built on Stellar" x402 Facilitator (OpenZeppelin Channels) verifies, settles, and sponsors fees over USDC — the agent-to-service rail we had on the roadmap, already live.

450K ACCOUNTS → SPONSORED RESERVES

### Migration without friction

Sponsored reserves let Xeni cover the base reserve and USDC trustline for every migrated wallet, so no user needs to hold XLM to be on Stellar.

Economics: ~5s finality, ~$0.00001 fees, USDC as a first-class asset, and SEP-24 / SEP-31 fiat anchors.

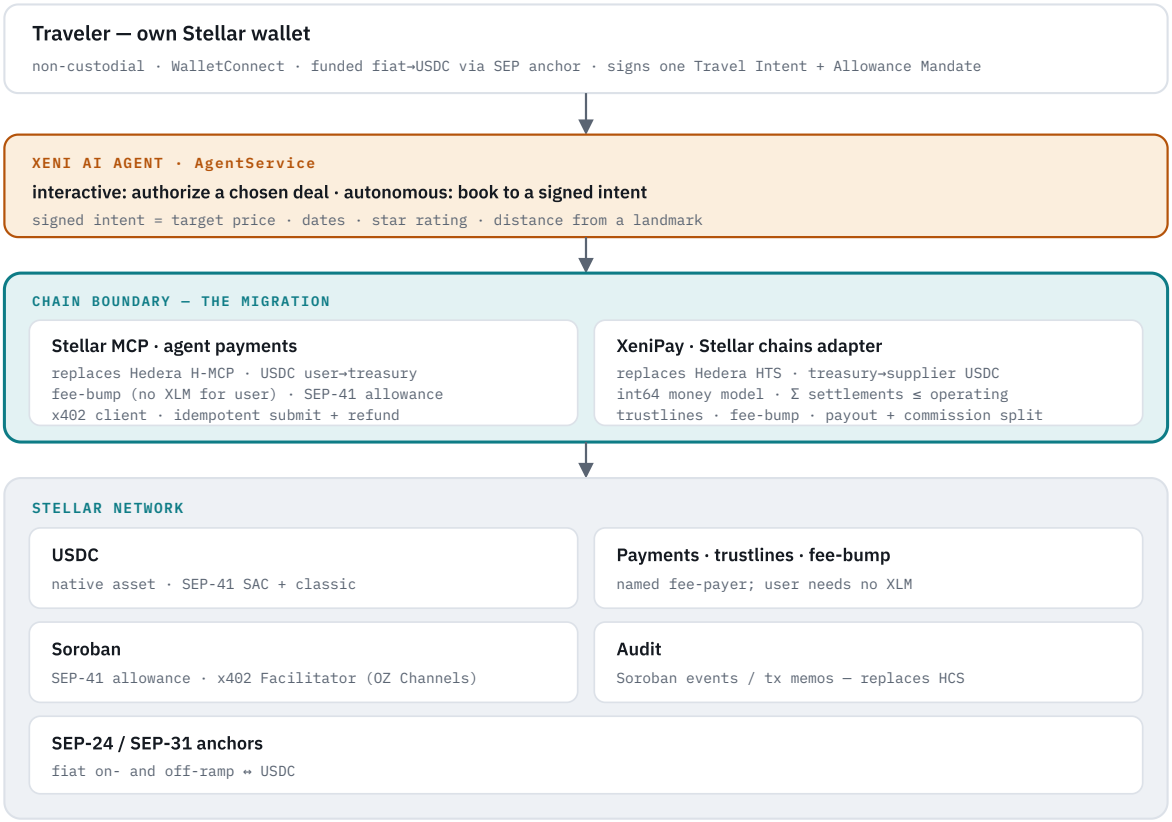
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3 · TARGET ARCHITECTURE

## Swap the chain boundary, keep everything above it

Two clean seams carry all on-chain movement: the **MCP boundary** in AgentService (user→treasury booking payments) and **XeniPay's chains adapter** (treasury→supplier

settlement). The migration replaces what sits beneath each — the agent's state machine, allowance guards, audit envelope, refund chain, and money model stay as they are.




Money path: traveler wallet → booking payment (Stellar MCP) → Xeni treasury → settlement (chains adapter) → suppliers; refunds run treasury → wallet back through the same MCP. The two teal boundaries are the migration's build; everything above stays.

## 3.1 · Components

**Traveler wallet.** The traveler's own non-custodial Stellar wallet, connected over WalletConnect and funded fiat→USDC through a SEP-24/31 anchor. Signs one Travel Intent + Allowance Mandate.

**Xeni AI Agent (AgentService).** Owns the intent state machine, allowance guards, audit envelope, and refund chain. Two modes: *interactive* (the traveler authorizes a chosen deal) and *autonomous* (the agent books to a signed intent = target price · dates · star rating · distance from a landmark). Unchanged by the migration.

**Stellar MCP — agent payments (migration work).** Replaces the Hedera H-MCP. Builds/submits USDC transfers `user→treasury`, fee-bump named-payer so the user holds no XLM, the Soroban SEP-41 allowance for autonomous spend, the x402 client, idempotent submit, and the hardened refund path (bounded retry, duplicate-tx recovery via Horizon, ops reprocess). Emits the audit envelope to Soroban events / tx memos.

**XeniPay Stellar chains adapter — settlement (migration work).** Replaces Hedera HTS. `treasury→supplier` / commission USDC payouts and trustlines, behind the `int64` money model with the `Σ settlements ≤ operating` conservation rule.

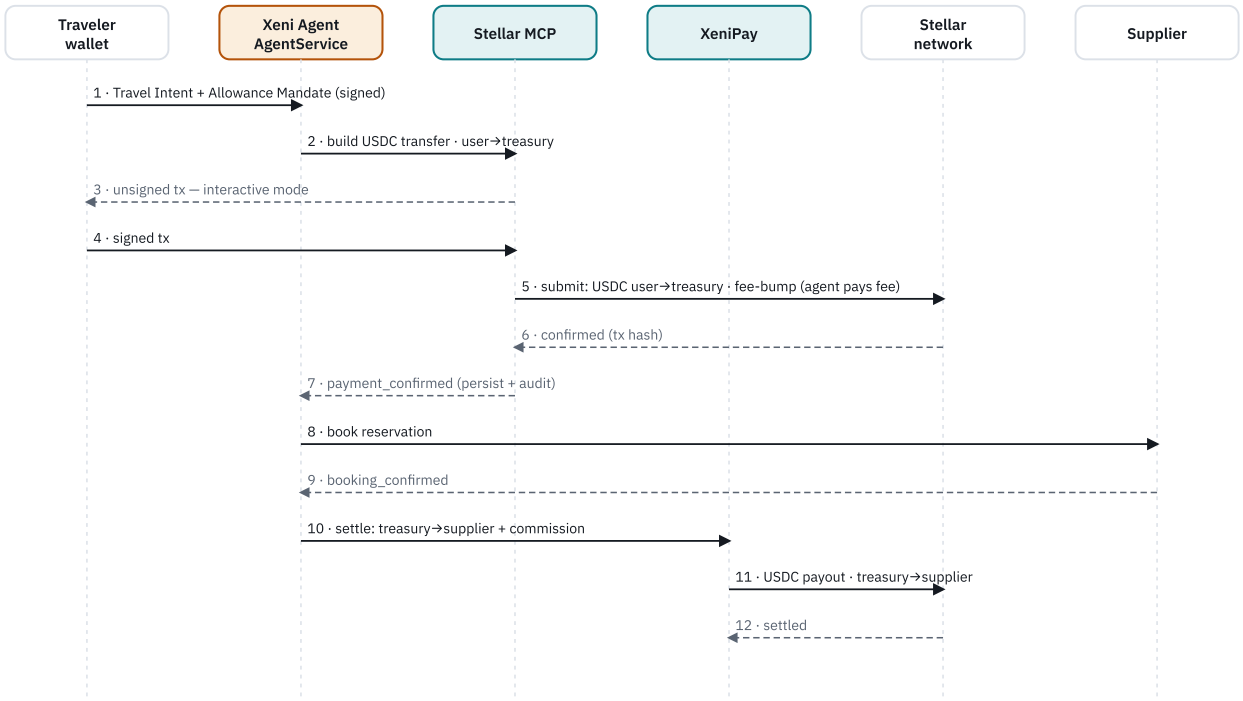
**Stellar network.** USDC (SEP-41 SAC + classic), Payments / trustlines / fee-bump, Soroban (SEP-41 allowance + x402 Facilitator), audit via Soroban events / tx memos, and SEP-24 / SEP-31 anchors.

### 3.2 · Migration seam

The clean swap points are the **MCP boundary** (agent payments) and `chains.BlockchainClient` (settlement). Both are already language-agnostic over the wire, so the migration is a new adapter behind each seam — not a rewrite of anything above it. The money invariants (§1) and the audit-envelope contract are the shared pieces that stay constant across the swap.

### 3.3 · End-to-end sequence — sub-system interactions

How the Xeni sub-systems interact to take a booking from a signed intent through payment, booking, and supplier settlement. The interactive mode is shown; the autonomous variant and the refund path are noted below.




Solid arrows are requests/commands; dashed are responses. **Autonomous mode:** steps 3–4 collapse into a one-time Soroban SEP-41 allowance signed at intent time — the agent then pays via transfer_from with no wallet round-trip. **Refund** (cancel/failure): XeniPay returns treasury → wallet through the same rail, audited and idempotent.

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Xeni — autonomous intent-based travel, settling on Stellar. Technical Architecture Document. Companion to the Project Plan. Not affiliated with or endorsed by the Stellar Development Foundation.